WORKSHEET · SCRATCH (GR 6–8) BEGINNER: CODE FOUNDATIONS

Lesson 5: Forever Loops and Conditions

Module 2 | 45 minutes | Conditional loops | Grades 6–8 · ages 11–14

Name: _____ Class: _____ Date: _____

Practise what you learned today. Read each task carefully.

1. Key words

Match each word to what it means. Write the letter.

a. selection ___ A. Checking again and again.
b. condition ___ B. The loop that runs a game every frame.
c. game loop ___ C. A true/false question.
d. polling ___ D. Choosing what to do with if-then.

2. Plan your code

Before you build it: Wrap-around car: forever, change x by 6. If x position > 230, set x to -230, so the Car reappears on the left. Write or draw the blocks you will use.

3. Debug the ball

This ball should bounce around forever, but it stops at the first wall. What's wrong? Circle one.



a. A. The blocks are not inside a forever loop
b. B. It needs a bigger number
c. C. It needs a wait block

4. Plan your code

Before you build it: Bumper ball: the Ball moves 8 steps forever and bounces on edges. If it touches the Bumper, it changes its colour effect by 25. Write or draw the blocks you will use.

5. Explain your thinking

Describe the game loop of a game you know: what does it check every frame, and what does it do when each condition is true?

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

- ☐ I can use a forever loop to keep a program running.
- ☐ I can use if-then to make decisions.
- ☐ I can explain a game loop.